PART-16

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Probability Distribution :-

- Random Variable
  - ↳ Discrete → finite
  - ↳ Continuous → Interval form (Uncountable)

- **probability mass fn (PMF) :-**
  - ↳ Discrete Random variable

- **probability Density fn (PDF) :-**
  - ↳ continuous Random variable

- **PMF :-**

when tossing four fair coins

- 0 head (TTTT)
- 1 head (HTTT, THTT, THHT, TTTH)

Total sample space = {1, 0}

$P(0) = \frac{1}{16}$, $P(1) = \frac{4}{16}$, $P(2) = \frac{6}{16}$ these all are probability of head.

→ we represent PMF By using "Bar Chart".

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⊛ PDF :-

PDF is represented / plotted By Histogram plot. (interval residu)

→ PDF based on interval data (continuous)

⊛ $$\boxed{PDF = \int_a^b f(x)\,dx}$$

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